

Environmental cadmium and breast cancer risk

.

Breast cancer is the most prevalent women's cancer, with an age-adjusted incidence of 122.9 per 100,000 US women. Cadmium, a ubiquitous carcinogenic pollutant with multiple biological effects, has been reported to be associated with breast cancer in one US regional case-control study. We examined the association of breast cancer with urinary cadmium (UCd), in a case-control sample of women living on Long Island (LI), NY (100 with breast cancer and 98 without), a region with an especially high rate of breast cancer (142.7 per 100,000 in Suffolk County) and in a representative sample of US women (NHANES 1999-2008, 92 with breast cancer and 2,884 without). In a multivariable logistic model, both samples showed a significant trend for increased odds of breast cancer across increasing UCd quartiles (NHANES, $p=0.039$ and LI, $p=0.023$). Compared to those in the lowest quartile, LI women in the highest quartile had increased risk for breast cancer (OR=2.69; 95% CI=1.07, 6.78) and US women in the two highest quartiles had increased risk (OR=2.50; 95% CI=1.11, 5.63 and OR=2.22; 95% CI=.89, 5.52, respectively). Further research is warranted on the impact of environmental cadmium on breast cancer risk in specific populations and on identifying the underlying molecular mechanisms.